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Taming the non-linearity: An iterative conceptual routing model for improving flood peak prediction

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ABSTRACT

The magnitude and timing of flood peaks are strongly influenced by the non-linear relationship between flow velocity and discharge. Traditional routing models struggle to account for the variation in flow velocity, particularly with time, due to computational constraints. This study proposes a novel Iterative Routing Model (IRM) that updates flow velocity as a function of streamflow magnitude. The IRM was applied to seven gauging stations in the Godavari River Basin, India. It outperformed two other models used in this study in simulating peak discharge and timing, with the lowest average absolute deviations of 29.98 % and 0.2 days, respectively. The IRM achieved the highest median NSE (0.78) and KGE (0.79) values across all stations. Moreover, the calibrated Manning's roughness from the proposed model appears more realistic compared to that given by other models. Overall, our findings highlight the potential of the proposed model to improve flood peak predictions in large river basins.

1. Introduction

Floods can cause extensive damage to life and property, underscoring the importance of precisely predicting their magnitude and timing. Hydrological models are generally used to predict different streamflow dynamics. The precise formulation of the runoff routing process in these models can help effectively capture flood peaks (Hamitouche et al., 2025; Markovic and Koch, 2015; Niu et al., 2017; Solanki et al., 2025; Zhao et al., 2017). Hydrodynamic routing models grounded in first principles numerically solve the shallow water equations to simulate the movement of runoff toward the basin outlet. However, their high data requirements and computational demands limit their applicability for larger spatial domains (Song et al., 2025; Teng et al., 2017; Unnithan et al., 2024).

Conceptual travel-time-based routing models (Clark, 1945; Nash, 1957; Rodríguez-Iturbe et al., 1982; Rodríguez-Iturbe and Valdés, 1979) derived from the unit-hydrograph theory (Sherman, 1932) are generally employed for routing the runoff in large basins. The concept of the time-area diagram proposed by Clark (1945), which represents the travel-time distribution of runoff from different parts of the basin to the outlet, provided a physical basis for the unit-hydrograph theory. Kirkby (1976) established a relationship between the streamflow response of a basin and its channel network morphology, providing the foundation for the development of Geomorphological Instantaneous Unit Hydrographs (Biswal and Singh, 2017; Gupta et al., 1980; Kumar et al., 2007; Rodríguez-Iturbe and Valdés, 1979; Snell and Sivapalan, 1994). Maidment et al. (1996) suggested a distributed routing framework that estimates unit hydrographs for each grid cell and simulates the streamflow at the basin outlet as the sum of responses from each cell. Rigon et al. (2016) provided a comprehensive review of various travel-time-based GIUH frameworks.

Travel time-based GIUH formulations offer a means to incorporate the influence of the spatial configuration of the drainage network in modelling the basin's streamflow response (Gad, 2014; Kadu and Biswal, 2024; Li et al., 2020; Naden, 1992). However, similar to the classical unit hydrograph theory, the earliest GIUH-based approaches were developed under the assumption that the streamflow response is linear and time-invariant. This is in contrast to extensive empirical and theoretical evidence indicating that streamflow response is inherently non-linear and strongly influenced by temporal variability in flow velocity (Basso et al., 2023; Furey and Gupta, 2007; Leopold and Maddock Jr, 1953; Meyersohn, 2016; Rodríguez-Iturbe et al., 1992). Leopold and

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Maddock Jr. (1953) observed that the flow velocity (v) varies with the flow magnitude (Q), following a power-law, $v \propto Q^m$, where m is a constant that varies between 0.3 and 0.5. The value of m can be derived to be equal to 0.4 for a wide rectangular channel based on the Manning's equation (Manning, 1891), and the proportionality constant can be expressed as: $c = \frac{s^{0.3}}{n^{0.6} b^{0.4}}$ (Muzik, 1996), where s is the bed slope, n is the roughness coefficient, and b is the channel width. Therefore, the power-law relationship between v and Q can be expressed as:

$$v(t) = c \cdot Q(t)^{0.4} \tag{1}$$

In contrast to the substantial temporal variability observed in flow velocity, its spatial variation was found to be negligible (Leopold and Maddock Jr, 1953). This observation was further supported by the theoretical justification provided by Rodríguez-Iturbe et al. (1992), who demonstrated that a spatially uniform flow velocity minimises energy expenditure per unit channel length within the drainage network. Evangelista et al. (2023) observed that flow velocity scales with drainage area in a power-law relationship, with an exponent close to 0.1, which had been reported by earlier studies (Jowett, 1998; Leopold and Maddock Jr, 1953). The exponent of 0.1 implies that flow velocity remains fairly constant in space when compared to its significant variation over time. Leopold and Maddock (1953) explained that the upstream reaches with steep slopes and high roughness tend to have shallow flows, while downstream reaches exhibit deeper flows, low roughness and gentle slopes. Since flow velocity in Manning's equation is jointly influenced by slope, roughness, and hydraulic radius, these opposing trends often compensate for one another, resulting in fairly constant velocity in space. Arora et al. (2001) showed that a travel time-based framework incorporating spatial variability in flow velocity while assuming it to remain constant over time fails to adequately represent streamflow dynamics, thereby supporting the previously discussed experimental and theoretical insights.

Recent travel-time-based frameworks (Du et al., 2009; Paul et al., 2018; Bunster et al., 2019) assumed temporal and spatial flow velocity variation to address the limitations of earlier travel-time-based approaches. Du et al. (2009) developed a routing framework that allows flow velocity to vary in time and space based on an approximation of Manning's formula (Kouwen et al., 1993). However, they ignored the influence of upstream runoff contribution on the flow velocity in downstream grid cells, which may result in the underestimation of the peak flows (Bunster et al., 2019). Paul et al. (2018) extended this framework by incorporating a cell-to-cell routing scheme (López-Vicente et al., 2014; Mizukami et al., 2016; Piccolroaz et al., 2016; Terink et al., 2015) within a large-scale hydrological model. However, using a distributed roughness coefficient across grid cells led to a significantly high computational runtime in the order of tens of hours. Bunster et al. (2019) proposed a routing framework that considers the dynamic upstream contribution (DUC) of runoff to each grid cell. The DUC model framework provided a more accurate estimation of peak flows than that proposed by Du et al. (2009); however, the model ignores the influence of runoff generated within the grid cell while updating its flow velocity. Furthermore, the framework was evaluated in a small watershed with an area of 90 km^2, leaving its applicability to larger basins uncertain.

In the present study, we propose a parsimonious iterative routing model (IRM) based on Eq. (1) to incorporate temporal variation in the flow velocity while assuming it remains spatially constant. The model provides a simplified yet robust alternative to data-intensive hydrodynamic models, retaining computational efficiency while improving peak-flow prediction accuracy. The performance of the proposed IRM is evaluated in the Godavari River Basin in India and compared with two routing models: (i) the DUC model and (ii) a constant velocity model (CVM), which assumes flow velocity to be constant in both space and time for flood routing. This study aims to develop a simplified yet effective routing framework to enhance flood prediction accuracy by incorporating well-established principles on flow velocity variation.

2. Methods

The daily runoff corresponding to each grid cell, needed as input to the routing frameworks, was estimated by employing the Dynamic-Budyko (DB) model (Istalkar and Biswal, 2025). A brief description of the DB model is provided in section 2.1. Section 2.2 describes the proposed routing framework with the iterative procedure for updating the flow velocity. Section 2.3 provides a brief description of the routing models used for comparison with IRM. Section 2.4 details model calibration and performance metrics used to evaluate the model's performance. Section 2.5 gives steps to apply the routing methodology.

2.1. Runoff estimation using DB model

The DB model is a calibration-free, continuous rainfall-runoff model developed for the estimation of runoff in data-scarce regions. It is based on the Budyko framework (Budyko, 1948) that performs long-term partitioning of rainfall into potential evapotranspiration (PET) and runoff depending upon the dryness index of the basin. Biswal (2016) proposed a concept of instantaneous dryness index (φ) for the short-term partitioning of rainfall into runoff using the Budyko function: $f(\varphi) = 1 - \left(\varphi.tanh\left(\varphi^{-1}\right) \cdot (1 - exp^{-\varphi})\right)^{0.5}$. The instantaneous dryness index is defined as the ratio of the total functional rainfall (FW) to the total functional solar energy (FH) available at a time. The FW and FH depend upon the cumulative effect of antecedent rainfall and solar energy inputs, respectively, which is assumed to decay exponentially with time (Istalkar and Biswal, 2025). The value of FH and FW at any time is estimated as:

$$FW(t) = \int_0^t W(\tau) \cdot exp^{-0.02(t-\tau)} d\tau \tag{2}$$

$$FH(t) = \int_0^t H(\tau) \cdot exp^{-0.02(t-\tau)} d\tau \tag{3}$$

Where $W(t) = R(t) - PET(t)$ is the water available in the system after satisfying the PET demand, $H(t) = PET(t) - R(t)$ is the available solar energy after the actual evapotranspiration of available water, and τ is a dummy variable. The instantaneous dryness index at any time, t, is, therefore, expressed as:

$$\varphi(t) = \frac{FH(t)}{FW(t)} \tag{4}$$

The effective rainfall (ER) is then estimated as follows:

$$ER(t) = W(t) \cdot f(\varphi) \tag{5}$$

ER is assumed to decay over time following the decay function: $x(t) = x(0) \cdot (1 + 0.4t)^{-1}$, proposed by Biswal (2016). Therefore, the runoff depth is estimated as follows:

$$q_{DB}(t) = \int_{t-\tau}^{t} -\frac{\partial}{\partial t}\left(\frac{ER(\tau)}{1 + 0.4 \cdot (t - \tau)}\right) d\tau \tag{6}$$

2.2. Proposed Iterative Routing Model (IRM)

The proposed iterative routing model employs a power-law relationship (Eq. (1)) between flow velocity and streamflow magnitude to account for temporal variation in velocity. A detailed methodological flowchart of the proposed framework is presented in Fig. 1. The runoff contribution from each grid cell to the basin outlet is determined by computing the travel time ($T_t = l/v$), where l represents the downstream

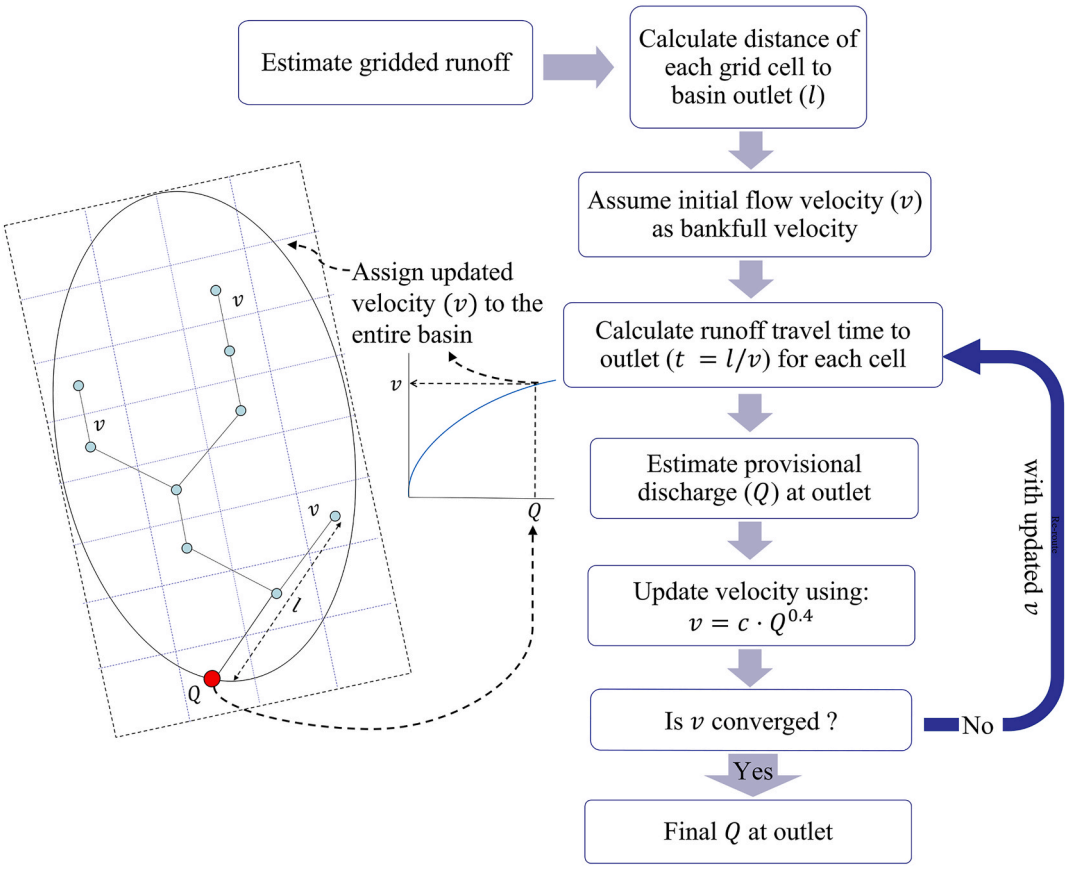


Fig. 1. Methodological flowchart of the iterative procedure for updating the flow velocity temporally.

distance of the grid cell from the outlet. Initially, the downstream distance of each Digital Elevation Model (DEM) pixel to the basin outlet along the stream network is computed using the D8 flow direction algorithm (O'Callaghan and Mark, 1984). Since the spatial resolution of the runoff grid is coarser than that of the DEM, the DEM pixel corresponding to the centre of each runoff grid cell is utilised to compute l. For a given time step, in the first iteration ($i = 1$), an initial estimate of travel time (T_i) is obtained by assuming the flow velocity (v_i) to be equivalent to the bankfull flow velocity (v_{bf}), which represents the velocity associated with the bankfull discharge (Leopold, 1994; Moody and Troutman, 2002). Bankfull discharge is defined as the discharge having a return period of approximately 1.6 years (Leopold, 1994; Pickup and Warner, 1976; Williams, 1978). The bankfull flow velocity is estimated using Manning's equation (Manning, 1891), incorporating the bankfull width and depth of the highest-order stream obtained from the global river geometry dataset (Andreadis et al., 2013). For the main Godavari basin, with its outlet at Polavaram, the bankfull discharge and bankfull velocity were calculated to be 25,862 m^3/s and 2.9 m/s, respectively. Based on the above initial estimate of flow velocity, the grid cells contributing runoff to the basin outlet are identified, and the streamflow (Q_i) for the first iteration is computed as the sum of runoff from these contributing cells.

In the next iteration ($i = 2$), the flow velocity is updated based on Eq. (1), utilising the streamflow estimated in the previous iteration, i.e., $v_2 = cQ_1^{0.4}$. The value of m is taken to be equal to 0.4 (Bunster et al., 2019; Du et al., 2009; Paul et al., 2018), which is a fixed parameter, and c is calculated by calibrating the Manning's coefficient for the basin. The updated velocity is used to recalculate the travel time and the corresponding streamflow (Q_2). The procedure is iterated until the flow velocity converges to a stable value to obtain the final streamflow estimate for a given time step. This iterative approach is applied at each time step to account for the temporal variation in flow velocity based on the streamflow magnitude.

2.3. Routing models used for comparison with IRM

The performance of IRM was compared with that of two other models: (a) the Dynamic Upstream Contribution Model (DUC; Bunster et al., 2019) and (b) a constant velocity routing model (CVM) employing a spatially and temporally constant flow velocity (v_c). The DUC model is a spatially distributed conceptual channel routing framework that estimates the travel time of an effective rainfall pulse generated in a grid cell as a function of the effective rainfall magnitude accumulated from its upstream cells. This approach allows for the velocity to vary both spatially and temporally. In contrast, the CVM assumes a constant flow velocity across both space and time, as is implicitly assumed in most conceptual models such as HBV, PDM, and Gr4J (Bergström, 1992; Moore, 2007; Perrin et al., 2003). To ensure an objective comparison among the three routing models and to isolate the performance differences arising solely from their routing structures, the runoff generation input (from the DB model) was kept identical for all models.

2.4. Model calibration and performance evaluation

The IRM and DUC model, used in this study, each consists of a single free parameter, the Manning's roughness coefficient, which was manually calibrated using the Nash–Sutcliffe Efficiency (NSE; Nash and Sutcliffe, 1970) as the objective function. In the constant velocity model (CVM), the spatially and temporally constant flow velocity, v_c, serves as the free parameter, which was calibrated using the same objective function as above. The daily streamflow records from 1981 to 1999 were used for the model calibration, whereas the period from 2000 to 2018 was considered for the validation. In the case of IRM, the updated

velocity at the outlet of the main basin was used to route flow throughout the entire basin, including the six sub-basins. Consequently, no separate calibration was performed for the sub-basins. Model performance was evaluated using NSE, coefficient of determination (R^2), and Kling-Gupta Efficiency (KGE; Gupta et al., 2009).

$$NSE = 1 - \frac{\sum (Q_0 - Q_m)^2}{\sum (Q_0 - \overline{Q}_0)^2} \tag{7}$$

$$R^2 = \left(\frac{\sum (Q_0 - \overline{Q}_0)(Q_m - \overline{Q}_m)}{\sqrt{\sum (Q_0 - \overline{Q}_0)^2 \sum (Q_m - \overline{Q}_m)^2}} \right)^2 \tag{8}$$

$$KGE = 1 - \sqrt{(r-1)^2 + (\alpha - 1)^2 + (\beta - 1)^2} \tag{9}$$

Where, Q_0 and Q_m are the observed and modelled discharges, respectively, while $\overline{Q}_0$ and $\overline{Q}_m$ represent the mean, r is the Pearson correlation coefficient, α is the ratio of standard deviations, while β is the bias ratio.

In this study, we classified a peak-flow event as a flood event if its magnitude exceeded the bankfull discharge. The models were evaluated to capture the magnitude and timing of the flood peaks using the Relative Error (RE) and the Time Error (T_e), given as:

$$RE = \frac{Q_{fm} - Q_{fo}}{Q_{fo}} \times 100 \tag{10}$$

$$T_e = t_{fm} - t_{fo} \tag{11}$$

Where Q_{fm} and Q_{fo} are the modelled and observed flood-peak magnitudes, respectively, while t_{fm} and t_{fo} are the modelled and observed times to the flood peak, respectively.

The Average Absolute Relative Error ($AARE$), the Average Absolute Error in their Timing (AAT_e), and the Peak Percentile Threshold Statistics (PPTS; Lohani et al., 2014) corresponding to the streamflow above the 95th percentile ($PPTS_{95}$) were calculated to determine the overall model performance during flood peaks.

$$AARE = \frac{1}{p} \sum_{i=1}^{p} |RE_i| \tag{12}$$

$$AAT_e = \frac{1}{p} \sum_{i=1}^{p} |T_{ei}| \tag{13}$$

$$PPTS_{95} = \frac{1}{n} \sum_{i=1}^{n} |RE_i| \tag{14}$$

Where p is the number of flood-peak events and n is the number of observed streamflow values above the 95th percentile.

2.5. Implementation of IRM

The methodology is implemented in MATLAB and Python, and the complete source code is available at https://zenodo.org/records/16937217. The following are the steps to implement the IRM.

(a) Inputs required to run IRM:
- $R[i,t]$: Runoff (m^3/s) at grid cell i and day t (see Section 2.1).
- $L[i]$: Downstream flow distance (m) from grid cell i to the basin outlet (derived from DEM via D8)
- v_{bf}: Bankfull velocity (m/s) to start the iterative process, estimated via Manning's equation using bankfull geometry (Andreadis et al., 2013; see Section 2.2).
- n: Manning's roughness coefficient, the only parameter to be calibrated (see Section 2.4).
- Δt: Routing time step (s).

(b) Pre-processing:
1. Derive flow direction and flow accumulation from the DEM using D8.
2. Delineate the stream network and compute downstream distance from each DEM pixel to the basin outlet along the network using the module '1_downstream_distance.m'.
3. Map the DEM pixel corresponding to the centroid of each runoff grid cell to obtain its distance to the outlet $L[i]$ at the runoff grid resolution. This can be done using the module '2_centroid_code.m'.

(c) Iterative procedure to update the flow velocity (use the module '3_IRM_code.py'). At each time step t:
1. Initialize iteration counter $itr = 1$, set $v_{itr} = v_{bf}$.
2. For each runoff grid cell $R[i]$, compute travel time to reach the outlet, $T_i = \frac{L[i]}{v_{iter}}$.
3. Identify contributing cells with $T_i \leq \Delta t$.
4. Compute provisional discharge at the outlet by summing the runoff from the contributing grid cells: $Q_{itr} = \sum R[i]$
5. Update velocity using Eq. (1): $v_{new} = c \cdot Q_{itr}^{0.4}$.
6. Convergence check: if $v_{new} = v_{itr}$ i.e., two successive iterations equal yield equal velocity, accept Q_{itr} as the discharge for the current time step; else set $v_{iter} = v_{new}$ and repeat Steps 2–6.

3. Study area and data

We tested the performance of our proposed IRM in the Godavari basin in India (Fig. 2), including its six sample sub-basins. The basin is the second largest in the country, having a drainage area of 312,812 km^2. The large size of the basin is particularly suitable for this study, as the flood waves propagate over multi-day periods, making it appropriate to analyze variations in peak timing and magnitude at a daily time step. The river originates from the Western Ghats in the state of Maharashtra and drains eastwards to meet the Bay of Bengal. The outlet of the basin is situated near Polavaram town in the state of Andhra Pradesh. The outlets of the six sub-basins are located near the towns of Ashti, Bamani, Tekra, Perur, Chindnar, and Pathgudem, with corresponding drainage areas of 50,990 km^2, 46,020 km^2, 108,780 km^2, 268,200 km^2, 17,270 km^2, and 40,000 km^2, respectively. The basin receives most of its rainfall during the Indian Summer Monsoon (ISM) between June and October, with an annual average of 1110 mm.

The 0.25° × 0.25° gridded daily precipitation and temperature datasets provided by the India Meteorological Department (IMD) were used as inputs to the DB model to estimate effective rainfall. The average daily observed streamflow time series was obtained from the India-Water Resources Information System (IndiaWRIS). Digital Elevation Model (DEM) data at a 30 m spatial resolution provided by the Shuttle Radar Topographic Mission (SRTM) was used for the delineation of basins and to obtain the corresponding stream networks. Bankfull width and the corresponding flow depth for the stream network were derived from the global river geometry dataset developed by Andreadis et al. (2013).

4. Results

The calibrated value of Manning's roughness coefficient for the proposed IRM was found to be 0.022, while for the DUC model, it was 0.012. The calibrated constant velocity for the CVM was determined to be 3.1 m/s. As previously mentioned, the IRM updates the flow velocity depending upon the streamflow magnitude at each time step through an iterative procedure until it converges to a stable value. Fig. 3 illustrates the convergence profiles for the three highest flood peaks observed in the main basin. These profiles demonstrate the progressive increase in flow velocity after every iteration, beginning with the initially assumed bankfull velocity. It is evident that flow velocity varied significantly across different flood events, deviating from the constant value of calibrated velocity for the CVM.

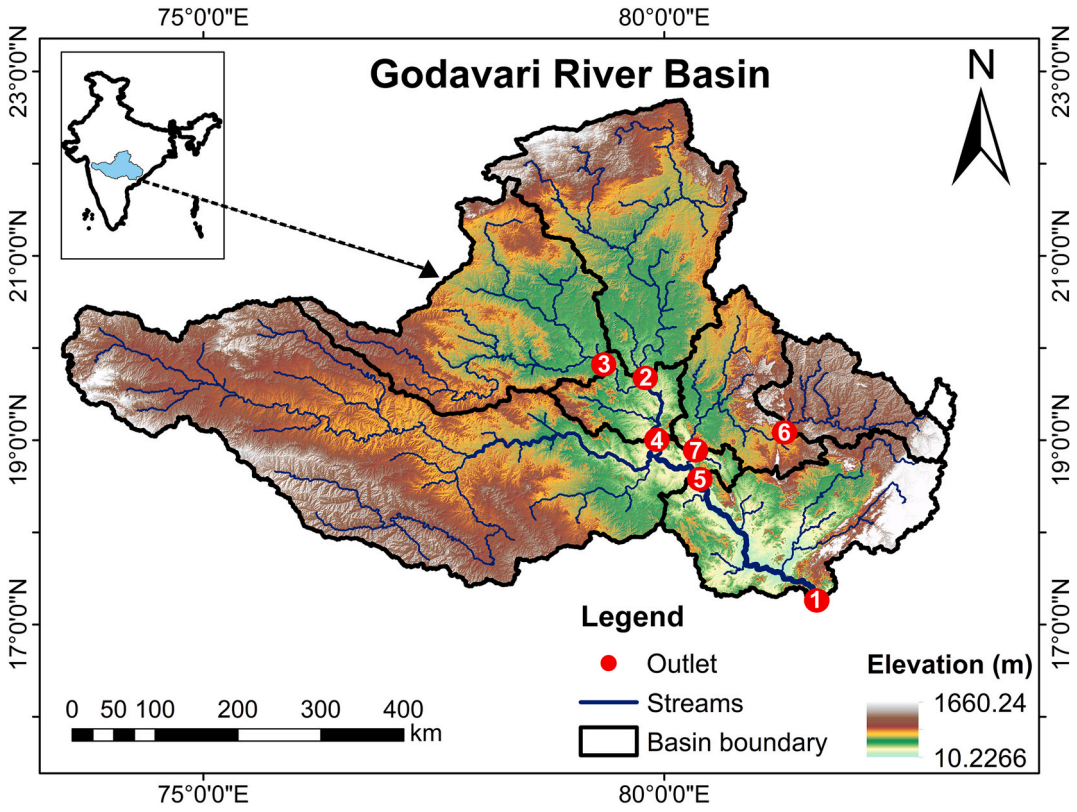


Fig. 2. Godavari River Basin and its sub-basins with their outlets at (1) Polavaram, (2) Ashti, (3) Bamni, (4) Tekra, (5) Perur, (6) Chindnar, (7) Pathgudem. The inset map shows the location of the Godavari Basin in India.

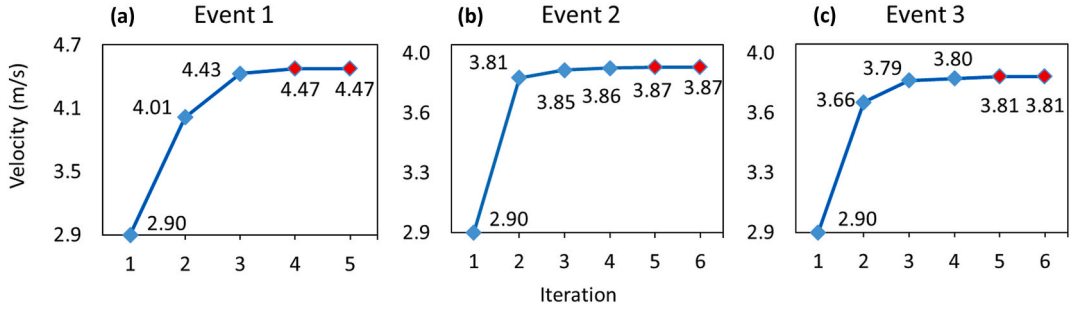


Fig. 3. Velocity convergence profile from the proposed iterative routing model corresponding to the flood peak observed on: (a) August 7, 2006, (b) September 21, 2006, and (c) August 10, 2007. The iterative procedure generally takes five to six iterations to converge from the initially assumed bankfull velocity, based on the streamflow magnitude on that day.

Fig. 4a illustrates the performance of the IRM in simulating the entire streamflow time series for the main Godavari basin, having the outlet at Polavaram. The R^2 was found to be 0.89, the NSE was 0.81, and the KGE was 0.80, indicating reliable model accuracy in predicting the entire streamflow time series. The R^2 corresponding to the DUC model and the CVM was found to be 0.79 and 0.84, respectively; NSE was found to be 0.73 and 0.74, respectively; and KGE was found to be 0.65 and 0.76, respectively.

Fig. 4b, c, and 4d illustrate the performance of IRM for the three highest flood peaks in the basin. In comparison to DUC and CVM, the IRM was found to capture both the peak magnitudes and the timing of the peaks precisely. For the highest flood peak (Fig. 4b) observed on August 7, 2006 with a magnitude of 52811.44 m^3/s, the flood peak simulated by the proposed IRM was 60363.1 m^3/s, whereas that simulated by DUC and CVM was 70187.02 m^3/s, and 57279.8 m^3/s, respectively. The relative error, *RE*, corresponding to the IRM, DUC, and CVM, was 14.29 %, 32.9 %, and 8.46 %, respectively. Furthermore, the time error, T_e, for IRM, DUC, and CVM were zero-day, -1-day, and 1-day, respectively, exhibiting accurate prediction of the peak timing. While it may appear that the CVM captured the peak magnitude better than the IRM, its performance dropped drastically in the subsequent flood peak events as the CVM did not account for the temporal variation in the flow velocity. For the second flood peak (Fig. 4c) observed on September 21, 2006 with a magnitude of 46823.79 m^3/s, the performance of IRM was found to be better than the other two models. The IRM simulated the peak magnitude as 45015.5 m^3/s with an *RE* of −3.86 %, whereas DUC and CVM reported an *RE* of 2.51 % and −17.99 %, respectively. On the contrary, IRM reported a T_e of zero days, and that reported by DUC and CVM was -1-day and 1-day, respectively. For the third flood peak (Fig. 4d) of magnitude 47248.88 m^3/s observed on August 10, 2007, the simulated peak by IRM was 43957.1 m^3/s with an *RE* of −6.96 %. The simulated peak by DUC and CVM was 47229.13 m^3/s and 33816.1 m^3/s, respectively and a corresponding *RE* of −0.04 % and −28.42 %, respectively. A T_e of zero days was reported by IRM, whereas that

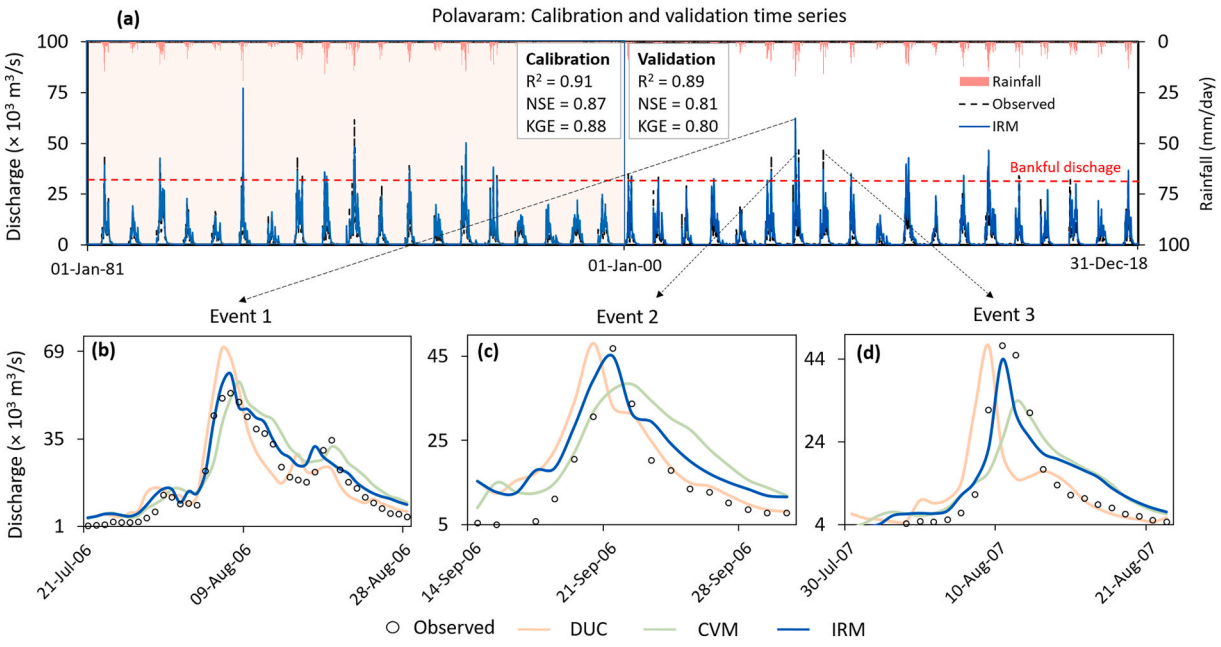


Fig. 4. (a) Performance of the proposed iterative routing model (IRM) during calibration and validation periods, (b), (c) and (d): Comparative hydrographs for peak events modelled by the IRM (blue), DUC (orange), and CVM (green). The IRM captures both the magnitude and timing of the flood peaks more accurately than the other two models.

reported by DUC and CVM was −1 day and 1 day, respectively. The above results suggest that the IRM can accurately capture the magnitude and timing of flood peaks.

Fig. 5 shows the top three flood peak events for each sub-basin. CVM was found to underestimate the flood peak magnitude, whereas DUC was found to overestimate the flood peak magnitude compared to the other two models in multiple instances (Fig. 5e, f, 5i, 5j, 5k, 5l, 5n, and 5o). In terms of predicting the timing of flood peaks, delayed peaks were simulated by CVM, while advanced peaks compared to the observed peaks were simulated by DUC. Across all sub-basins, IRM was found to perform better in terms of estimating both flood peak magnitude and its timing.

The highest flood peak event among all six sub-bains was observed in Perur (Fig. 5j) of magnitude 62889.13 m^3/s on August 15, 1986. The corresponding modelled flood peaks by the IRM, DUC, and CVM were 60383 m^3/s, 61137.59 m^3/s, and 57807.9 m^3/s, with an *RE* of −3.9 %, −2.78 %, and −8.07 %, respectively. Moreover, T_e was zero-day, -1-day and 1-day, respectively, exhibiting a better capture of flood peak by IRM than the other two models. The Perur and Tekra sub-basins, being the largest sub-basins, had observed continuous flood peak events at four instances (Fig. 5h, i, 5k, 5l). One such instance was observed in the Perur sub-basin (Fig. 5l), where two continuous peaks were seen. The highest observed peak magnitude was 51916.9 m^3/s. IRM, DUC, and CVM simulated the peak magnitude with an *RE* of −2.48 %, 10.15 %, and −19.64 %, respectively. T_e was found to be zero-day, zero-day, and 1-day by IRM, DUC, and CVM, respectively. The lowest *RE* and T_e reported for IRM shows the ability of the iterative framework to capture the flood peak magnitude and timing effectively.

Fig. 6 illustrates the scatter plot of observed vs. modelled flow corresponding to the three models across six sub-basins of the Godavari River basin. IRM reported a median R^2 of 0.79, whereas DUC and CVM reported a median R^2 of 0.75, and 0.77, respectively, exhibiting that IRM captured the streamflow variability better than the other routing models. Moreover, the median NSE reported by IRM, DUC, and CVM was 0.78, 0.72, and 0.76, respectively. In terms of KGE, the IRM consistently performed better than the other routing models across all sub-basins in capturing the streamflow variability with a median KGE of 0.79 compared to a median KGE of 0.72, and 0.76 reported by DUC and CVM, respectively.

Fig. 7a shows the distribution of *RE* for the IRM, DUC, and CVM for the flood peaks across the seven sub-basins. For the main basin, the median *RE*, corresponding to IRM, DUC, and CVM, was found to be −3.81 %, 5.5 % and −17.99 %, respectively. The distribution of T_e across all sub-basins is illustrated in Fig. 7b. The median T_e, averaged across all sub-basins, for the IRM was found to be zero days, whereas it was −0.5 days and 0.57 days for the DUC and CVM, respectively. For the uncertainty in T_e (shown by interquartile range), the DUC model reported the mean 25th percentile and 75th percentile of T_e to be equal to −0.71 days and −0.42 days, respectively, which shows that DUC tend to estimate the timing of flood peaks earlier than observed. The mean 25th percentile and mean 75th percentile T_e reported by CVM were 0 days and 1 day, respectively. IRM reported the lowest interquartile range when compared to the other two routing models across all sub-basins, with a mean 25th percentile and 75th percentile T_e of −0.08 days and 0.14 days, respectively. This exhibits the ability of IRM to estimate the timing of flood peaks with the lowest uncertainty compared to the other routing models.

For an aggregate model performance during peaks, we also analysed average absolute relative error (*AARE*), peak percentile threshold statistic ($PPTS_{95}$) and average absolute time error (AAT_e) shown in Fig. 8b, c, and 8d, respectively. IRM was found to perform better than the other two routing models, with IRM having the lowest median *AARE*. The median *AARE* by IRM was found to be 29.98 %, whereas, for DUC and CVM, it was found to be 31.19 %, and 32.62 %, respectively. Similarly, the median $PPTS_{95}$ reported by DUC, CVM, and IRM were 30.13 %, 28.16 %, and 23.64 %, respectively. With the least median $PPTS_{95}$ across all sub-basins, IRM was found to capture the top 5 percentile flow more accurately compared to the other two routing models. Moreover, IRM captured the peak timings with the least median AAT_e. The median AAT_e across all sub-basins were reported to be 1 day, 0.9 days, and 0.2 days by DUC, CVM and IRM, respectively. The above results exhibit a better performance in capturing the flood peak magnitude and timing by IRM than DUC and CVM.

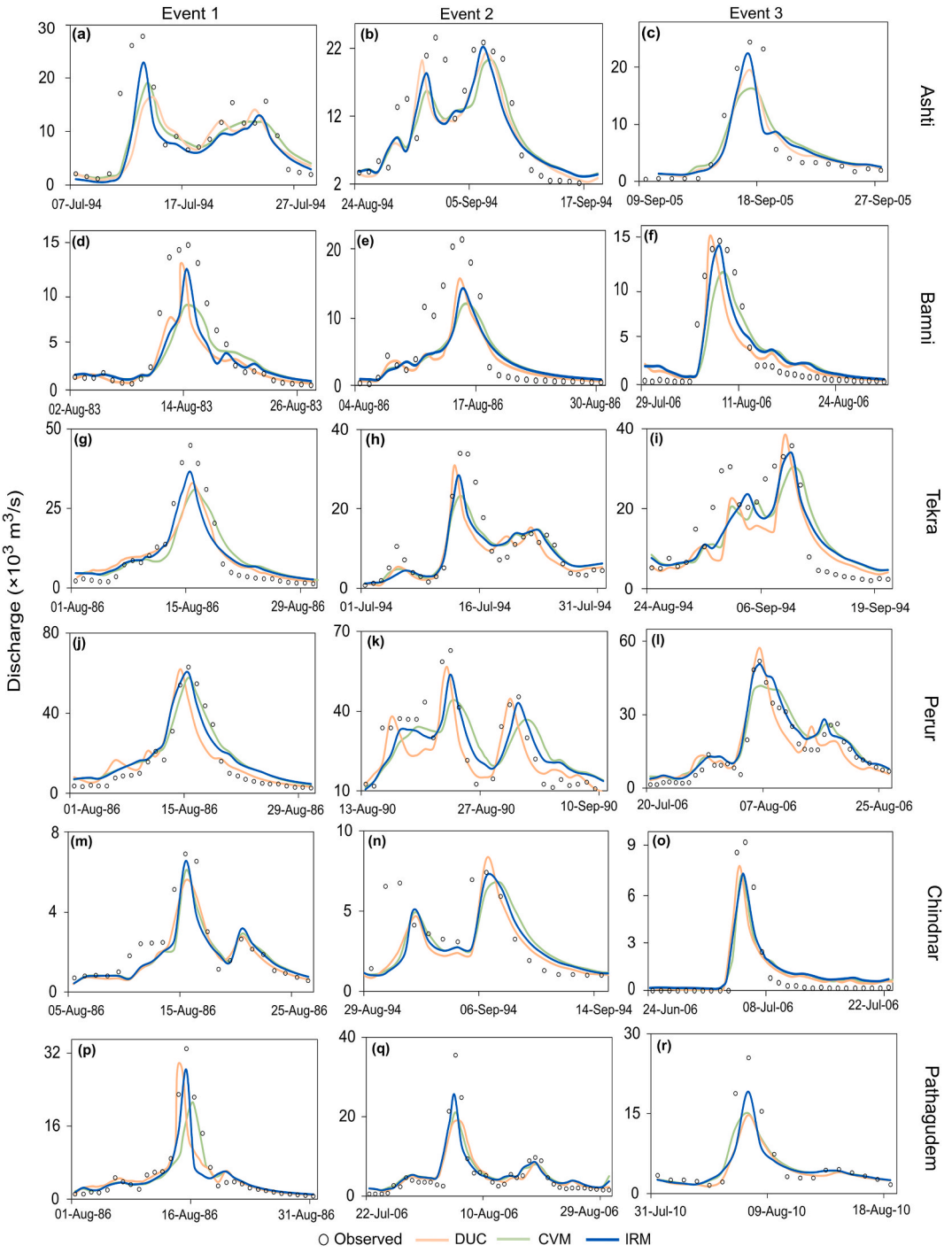


Fig. 5. Peak events simulated with the DUC, CVM, and the proposed IRM across all sub-basins. Whether for isolated single events or multiple continuous events, the proposed IRM (blue) captures both the observed (black circles) peak magnitude and timing more accurately than the DUC (orange) and CVM (green) models. The DUC tends to slightly overestimate and predict peaks earlier, whereas the CVM generally underestimates and delays peak occurrence across sub-basins and events.

5. Discussion

The calibrated value of Manning's roughness coefficient obtained for the IRM ($n = 0.022$) was more realistic compared to that obtained for the DUC model ($n = 0.011$). The roughness coefficient values typically range from 0.02 to 0.04 for natural river channels, whereas values between 0.011 and 0.018 are characteristic of paved concrete surfaces (Arcement and Schneider, 1989).

A low roughness coefficient was calibrated for the DUC model to compensate for the overestimation of travel time, which may have resulted from neglecting past accumulated rainfall within a given grid cell. In large basins, the accumulation of past rainfall can significantly reduce travel time, leading to a sharper rise in the hydrograph (Rinaldo et al., 2011). The IRM accounts for the influence of past rainfall within a grid cell through an iterative procedure. At each time step, after every iteration, the effect of rainfall from both upstream grid cells and the grid cell itself is incorporated into the travel time calculation. The DUC model also produced a steeper hydrograph, which can be attributed to a lower calibrated roughness coefficient, resulting in unrealistically low resistance to flow.

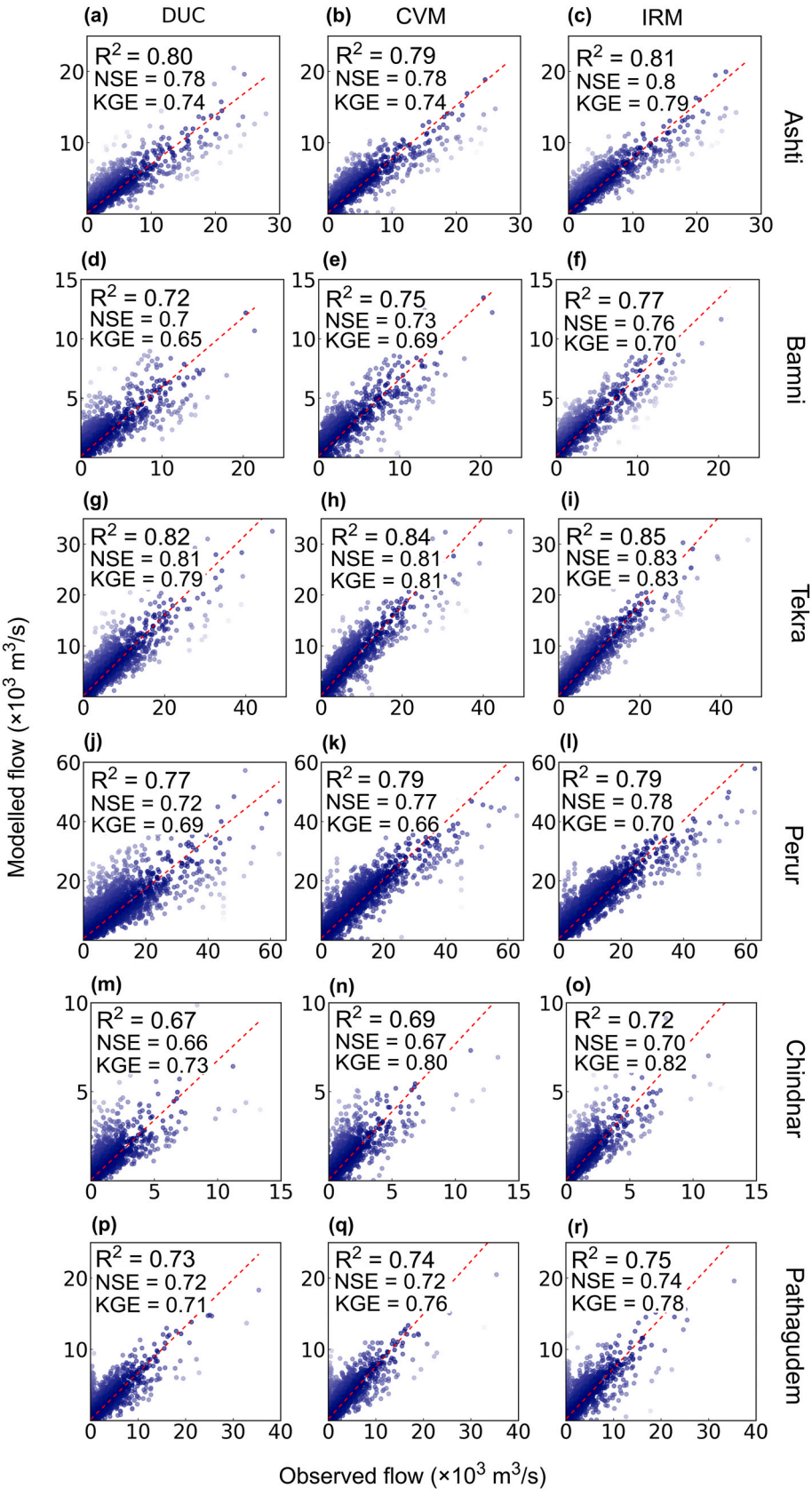


Fig. 6. Scatter plots showing the model performance in continuous modelling of streamflow by DUC (column 1), CVM (column 2), and IRM (column 3) with respect to observed flow.

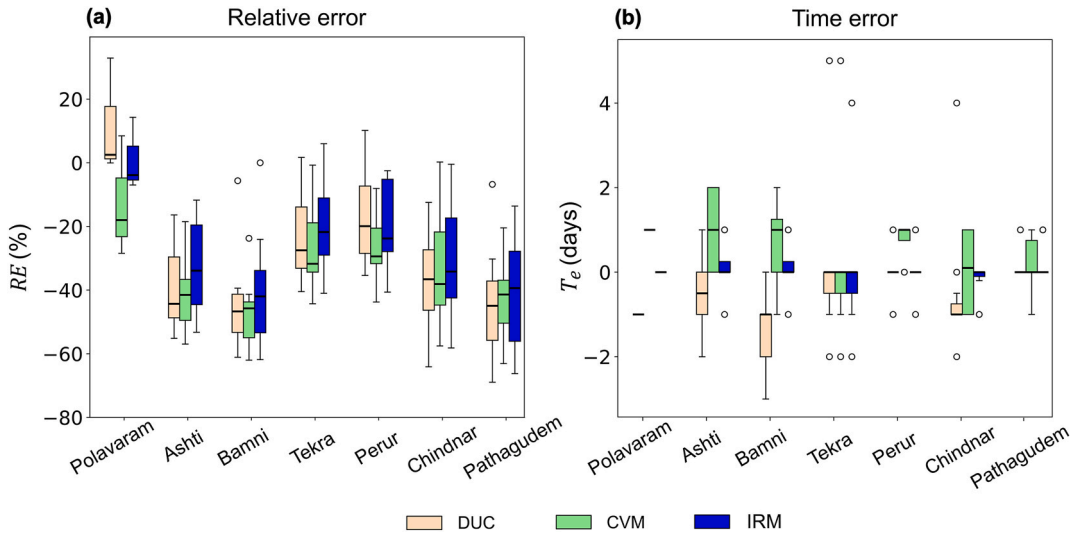


Fig. 7. (a) Distribution of relative error and (b) distribution of time error peak across peak events above bankfull discharge in all sub-basins. The box represents the interquartile range, with the lower and upper hinges marking the 25th and 75th percentiles, respectively, while the solid line within the box shows the median. Each box represents the distribution of *RE* or T_e reported by a model across all the observed flood peak events in a sub-basin during the simulation period. The IRM shows the lowest uncertainty in the estimation of peak magnitude and timing compared to the other two routing models across all subbasins.

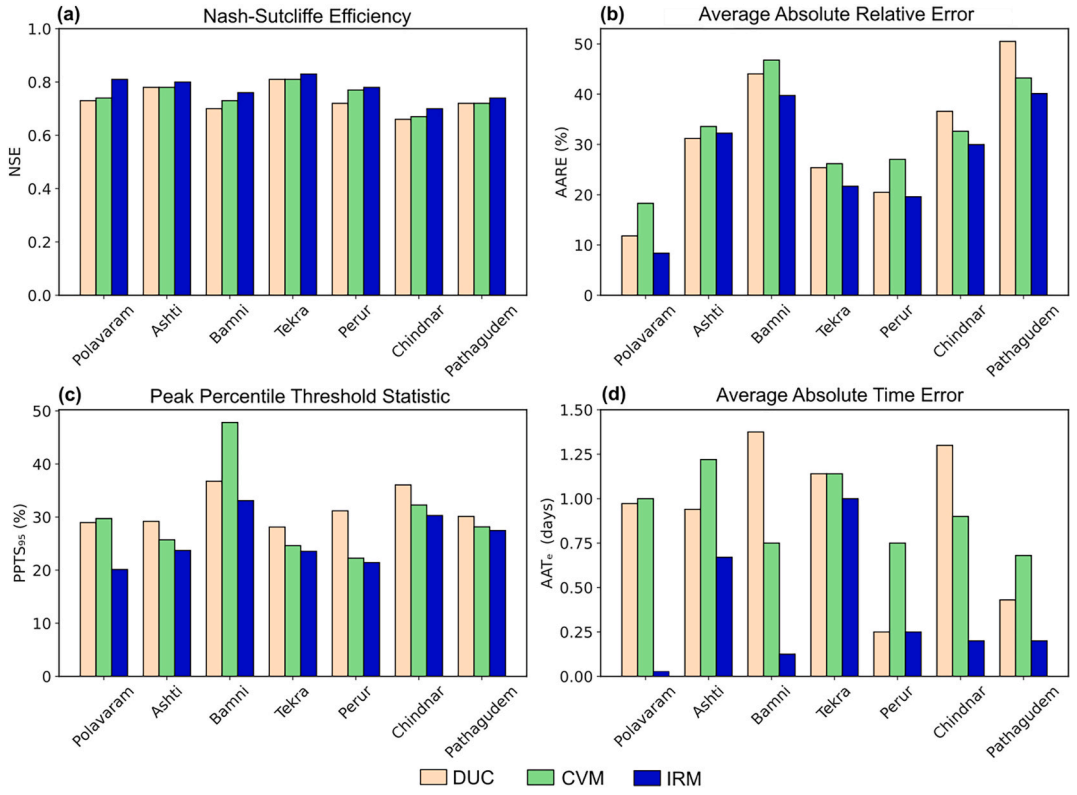


Fig. 8. Bar graphs comparing the model performance metrics reported by IRM with DUC and CVM across all sub-basins. (a) NSE; (b) Average absolute relative error (*AARE*); (c) Peak percentile threshold statistic ($PPTS_{95}$); and (d) Average absolute time error (AAT_e). The IRM (blue bar) demonstrates the least error in estimation of peak magnitude and timing across all subbasins.

The CVM resulted in lower and delayed flood peaks, particularly in the main basin and the larger sub-basins, which can be attributed to the higher flow velocity during flood events compared to the constant velocity calibrated for the entire flow period. During peak flow conditions, an additional wave celerity component increases the average flow velocity, leading to sharper peaks. As the hydrograph approaches its crest, the wave propagation rate increases with discharge at each time step (Beg et al., 2023). The IRM, which accounts for the increase in flow velocity with discharge magnitude, can thus capture flood peaks more accurately compared to the CVM. These findings concur with those of Koussis and Mazi (2016) and Bharali and Misra (2022) who observed an improvement in the peak flow prediction by updating the routing parameters based on the discharge magnitude. Notably, the IRM exhibited reduced scatter during low-flow conditions (Fig. 6) compared to other

models across all sub-basins. This suggests that the IRM effectively updates flow velocity not only during high-flow events but also under low-flow conditions, enhancing its reliability across different hydrological scenarios.

In this study, to simulate daily streamflow over 38 years for each sub-basin on a system equipped with a 16-core Xeon Gold processor, the computational time required for DUC was approximately 7 h, CVM was 5 min, and IRM was 20 min. The structure and parameterisation of the IRM can thus be considered parsimonious, computationally efficient, yet realistic for larger basins compared to the other models.

To assess the implications of assuming a spatially constant velocity in IRM, we quantified the relative difference in discharge obtained under spatially constant velocity and under velocity varying with drainage area based on the area–velocity scaling relation ($v \propto A^{0.1}$; Costabile et al., 2024; Evangelista et al., 2023; Jowett, 1998; Leopold and Maddock Jr., 1953). The relative differences were estimated using a first-order Taylor approximation (Appendix A, Eq. A10). As summarized in Table 1, deviations were minor: approximately 0.5–0.7 % for medium-sized sub-basins (Ashti, Bamni, Pathagudem), about 1 % for the smallest sub-basin (Chindnar), and negligible (0.05 %) for the largest upstream sub-basin (Perur). These results confirm that the simplification of assuming spatially constant velocity introduces only a minute error in discharge estimation compared to the scaling law with an exponent of 0.1. Furthermore, a sensitivity analysis varying the velocity exponent m from 0.3 to 0.5 (Table 2) also showed that the best performance is at $m = 0.4$. The value of $m = 0.4$ is valid for wide rectangular channels, and most of the natural river channels can be reasonably approximated as wide rectangular channels (Singh et al., 2003).

The uncertainty in the input data may influence the model performance. In particular, the uncertainty in the river geometry data may impact the routing performance. The effect of uncertainty in channel slope and width estimation on model performance is shown in Fig. 9. The channel slope along the longest drainage path were estimated using three approaches: (a) exponential smoothening of the elevation profile (Unnithan et al., 2024), (b) elevation difference between the upstream and downstream endpoints, divided by the distance between the points, and (c) elevation difference between points at the 10 % and 85 % distance measured from the outlet along longest drainage path, divided by the distance between the points (Benson, 1959). The influence of uncertainty in the estimation of channel width on model performance was represented by considering the width estimated by (i) width-discharge regression model (Andreadis et al., 2013), (ii) its lower bound, and (iii) its upper bound corresponding to 95 % confidence interval. The analysis shows that uncertainties in slope and channel width may lead to different values of calibrated Manning's roughness coefficient. Notably, maximum NSE remained above 0.73 across different slope–width combinations, suggesting that the calibration of the roughness coefficient effectively compensates for these uncertainties.

Since the analysis is conducted at a daily temporal resolution, numerical dispersion in flow may dominate actual dispersion, rendering the explicit incorporation of a dispersion equation unnecessary in the current formulation. However, accounting for streamflow dispersion may be essential for accurately simulating wave attenuation at finer temporal resolutions. Future applications of the IRM using sub-daily rainfall and streamflow observations would help in better evaluating its performance during the flash flood events. Additionally, incorporating the influence of snow-melt, channel-storage, and backwater-effect in the model structure may further enhance prediction accuracy. While hillslope routing was considered less critical for the large basins, future developments may adapt the framework for smaller catchments by integrating a hillslope routing component.

Table 1
Sub-basin-wise average discharge deviations (%) resulting from the assumption of spatially constant velocity in IRM, relative to locally scaled velocities following the area-velocity scaling law ($v \propto A^{0.1}$).

Sub-basin	Average discharge deviation (%)
Ashti	0.55
Bamni	0.54
Tekra	0.33
Perur	0.05
Chindnar	1.05
Pathagudem	0.71

Table 2
Model performance (NSE) across sub-basins under different discharge–velocity exponents.

Sub-basin		Exponent				
		0.3	0.35	0.4	0.45	0.5
Polavaram	**NSE**	0.78	0.79	0.81	0.77	0.76
Ashti		0.81	0.81	0.81	0.80	0.80
Bamni		0.73	0.74	0.76	0.73	0.72
Tekra		0.81	0.83	0.83	0.83	0.82
Perur		0.77	0.77	0.78	0.78	0.78
Chindnar		0.69	0.69	0.70	0.68	0.67
Pathagudem		0.73	0.72	0.74	0.71	0.70

6. Conclusions

Flow velocity is a key factor influencing both the magnitude of peak flow and its timing. Although it is widely recognised that flow velocity varies significantly over time, most conceptual routing frameworks assume it to be constant, implicitly or explicitly. In this study, a novel conceptual flow routing model was proposed that incorporates the temporal variation in flow velocity while assuming it to remain spatially constant. The model iteratively updates flow velocity at each time step using an empirical power-law relationship between channel velocity and discharge. This approach addresses the limitations of oversimplified conceptual routing models, such as the unit hydrograph method, which assumes a linear streamflow response of the basin.

The proposed routing model was tested in the Godavari River basin in India, along with its seven sub-basins and its performance was compared with two existing conceptual routing models: (i) DUC, which incorporates both temporal and spatial variations in flow velocity, and (ii) a constant-velocity routing model (i.e., CVM) that assumes flow velocity to be uniform in both time and space. The results demonstrated that the proposed iterative routing model accurately captures both the magnitude and timing of peak flows compared to the other two frameworks. Furthermore, a more realistic calibrated Manning's coefficient was obtained for the proposed model, implying its structure to be more realistic compared to the other models. Overall, the study shows that updating velocity within a parsimonious conceptual runoff routing framework improves the representation of flood peaks in large basins.

CRediT authorship contribution statement

Ekant Sarkar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Akshay Kadu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis. **S.L. Kesav Unnithan:** Writing – review & editing, Visualization, Validation, Software, Methodology, Conceptualization. **Basudev Biswal:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: S. L. Kesav Unnithan has a patent #571718 granted by the Indian Patent

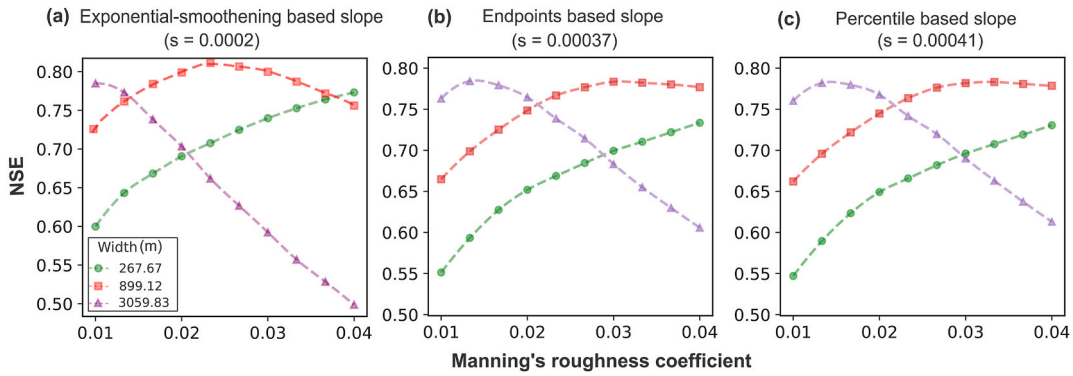


Fig. 9. Model performance (NSE) under different parameter combinations at the main basin. Slopes along the longest drainage path (LDP) were estimated using three methods: (a) slope from exponentially smoothed elevation profile of the LDP (Unnithan et al., 2024), (b) slope from elevation difference between the upstream and downstream endpoints of the LDP divided by path length, and (c) slope from elevation difference between points located at 10 % and 85 % of the LDP divided by the distance between the points (Benson, 1959). Three representative values for channel width were considered: the regression-based estimate (=899.12 m) and the lower (=267.67 m) and upper (3059.83 m) bounds of the 95 % confidence interval derived from the width–discharge relationship (Andreadis et al., 2013). Each subplot illustrates the sensitivity of model performance (NSE) to variations in Manning's roughness coefficient across the three channel widths, under the respective slope estimation method.

Office. Basudev Biswal has a patent # 571718 granted by the Indian Patent Office.

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Appendix A. Analytical derivation of discharge deviations under the spatially constant velocity assumption

For hydraulically consistent scaling,

$$v(A) = c\,A^{\alpha}, \alpha = 0.1 \tag{A1}$$

Now, velocity at the basin outlet,

$$v_0 = v(A_{out}) = cA_{out}^{\alpha} \tag{A2}$$

Let, a flow path has arclength coordinate $x \in [0, L]$, drainage area $A(x)$, and segment length dx.
Travel time under hydraulically consistent scaling, from Eq. (A1),

$$t_{scale} = \int_0^L \frac{dx}{v(A(x))} = \frac{1}{c}\int_0^L A(x)^{-\alpha} dx \tag{A3}$$

Also, travel time using velocity at the outlet, from Eq. (A2),

$$t_{IRM} = \int_0^L \frac{dx}{v_0} = \frac{L}{cA_{out}^{\alpha}} \tag{A4}$$

From Eqs. (A3) and (A4), the travel-time ratio is,

$$\frac{t_{IRM}}{t_{scale}} = \frac{A_{out}^{\alpha} L}{\int_0^L A(x)^{-\alpha} dx} \tag{A5}$$

For a path length L ,

$$\frac{t_{IRM}}{t_{scale}} = \left[\frac{A_i}{A_{out}}\right]^{\alpha}, A_i \text{is the drainage area for the lenth } L \tag{A6}$$

The relative travel-time deviation of IRM from the scaling law can be expressed as

$$\frac{t_{IRM} - t_{scale}}{t_{scale}} = \frac{\Delta t}{t_{scale}} = \left[\frac{A_i}{A_{out}}\right]^{\alpha} - 1 \tag{A7}$$

If $Q(t)$ denotes the hydrograph produced by IRM, then incorporating the travel-time deviation Δt implies that the shifted hydrograph is,

$$Q^{*}_{(t)} = Q(t + \Delta t) \tag{A8}$$

For small $|\Delta t|$ relative to the event duration, a first-order Taylor expansion gives

$$Q(t+\Delta t) \approx Q(t) + \frac{dQ}{dt}\Delta t \tag{A9}$$

and, hence the discharge deviation is given as

$$\frac{\delta Q}{Q} \approx \frac{1}{Q}\frac{dQ}{dt}\Delta t = \frac{d\ ln\ Q}{dt}\Delta t \tag{A10}$$

Data availability

The model codes are available at https://zenodo.org/records/16937217.

References

Andreadis, K.M., Schumann, G.J.-P., Pavelsky, T., 2013. A simple global river bankfull width and depth database. Water Resour. Res. 49, 7164–7168. https://doi.org/10.1002/wrcr.20440.

Arcement, G.J., Schneider, V.R., 1989. Guide for selecting Manning's roughness coefficients for natural channels and flood plains. U.S. Geological Survey Water-Supply Paper 2339, v, 38 p. doi:10.3133/wsp2339.

Arora, V., Seglenieks, F., Kouwen, N., Soulis, E., 2001. Scaling aspects of river flow routing. Hydrol. Process. 15, 461–477. https://doi.org/10.1002/hyp.161.

Basso, S., Merz, R., Tarasova, L., Miniussi, A., 2023. Extreme flooding controlled by stream network organization and flow regime. Nat. Geosci. 16, 339–343. https://doi.org/10.1038/s41561-023-01155-w.

Beg, M.N.A., Meselhe, E.A., Kim, D.H., Halgren, J., Wlostowski, A., Ogden, F.L., Flowers, T., 2023. Diffusive wave models for operational forecasting of channel routing at continental scale. J. Am. Water Resour. Assoc. 59, 257–280. https://doi.org/10.1111/1752-1688.13080.

Bergström, S., 1992. The HBV model–its Structure and Applications.

Benson, M.A., 1959. Channel-slope factor in flood-frequency analysis. J. Hydraul. Div. 85, 1–9. https://doi.org/10.1061/JYCEAJ.0000291.

Bharali, B., Misra, U.K., 2022. Numerical approach for channel flood routing in an ungauged basin: a case study in Kulsi River Basin, India. Water Conserv. Sci. Eng. 7, 389–404. https://doi.org/10.1007/s41101-022-00149-w.

Biswal, B., 2016. Dynamic hydrologic modeling using the zero-parameter Budyko model with instantaneous dryness index. Geophys. Res. Lett. 43, 9696–9703. https://doi.org/10.1002/2016GL070173.

Biswal, B., Singh, R., 2017. Incorporating channel network information in hydrologic response modelling: development of a model and inter-model comparison. Adv. Water Resour. 100, 168–182. https://doi.org/10.1016/j.advwatres.2016.12.015.

Budyko, M., 1948. Evaporation Under Natural Conditions, Gidrometeorizdat, Leningrad, English Translation by IPST. Jerusalem.

Bunster, T., Gironás, J., Niemann, J.D., 2019. On the influence of upstream flow contributions on the basin response function for hydrograph prediction. Water Resour. Res. 55, 4915–4935. https://doi.org/10.1029/2018WR024510.

Clark, C.O., 1945. Storage and the unit hydrograph. Trans. Am. Soc. Civ. Eng. 110, 1419–1446. https://doi.org/10.1061/TACEAT.0005800.

Costabile, P., Barbero, G., Nagy, E.D., Négyesi, K., Petaccia, G., Costanzo, C., 2024. Predictive capabilities, robustness and limitations of two event-based approaches for lag time estimation in heterogeneous watersheds. J. Hydrol. 642. https://doi.org/10.1016/j.jhydrol.2024.131814.

Du, J., Xie, H., Hu, Y., Xu, Y., Xu, C.Y., 2009. Development and testing of a new storm runoff routing approach based on time variant spatially distributed travel time method. J. Hydrol. 369, 44–54. https://doi.org/10.1016/j.jhydrol.2009.02.033.

Evangelista, G., Woods, R., Claps, P., 2023. Dimensional analysis of literature formulas to estimate the characteristic flood response time in ungauged basins: a velocity-based approach. J. Hydrol. 627, 130409. https://doi.org/10.1016/j.jhydrol.2023.130409.

Furey, P.R., Gupta, V.K., 2007. Diagnosing peak-discharge power laws observed in rainfall-runoff events in Goodwin Creek experimental watershed. Adv. Water Resour. 30, 2387–2399. https://doi.org/10.1016/j.advwatres.2007.05.014.

Gad, M.A., 2014. Flow Velocity and Travel Time Determination on Grid Basis Using Spatially Varied Hydraulic Radius 23, pp. 36–46. https://doi.org/10.3808/jei.201400259.

Gupta, V.K., Waymire, E., Wang, C.T., 1980. A representation of an instantaneous unit hydrograph from geomorphology. Water Resour. Res. 16, 855–862. https://doi.org/10.1029/WR016i005p00855.

Gupta, H.V., Kling, H., Yilmaz, K.K., Martinez, G.F., 2009. Decomposition of the mean squared error and NSE performance criteria: implications for improving hydrological modelling. Water Resour. Res. 45, W10403. https://doi.org/10.1029/2009WR008898.

Hamitouche, M., Fosser, G., Anav, A., He, C., Lin, T., 2025. Impact of Runoff Schemes on Global Flow Discharge : a Comprehensive Analysis Using the Noah-MP and CaMa-Flood Models, pp. 1221–1240.

Istalkar, P., Biswal, B., 2025. DBv2: an improved climate-centric calibration-free model for runoff-generation simulation. Hydrol. Sci. J. https://doi.org/10.1080/02626667.2025.2581263, 0.

Jowett, I.G., 1998. Hydraulic geometry of New Zealand rivers and its use as a preliminary method of habitat assessment. River Res. Appl. 14, 451–466. https://doi.org/10.1002/(sici)1099-1646(1998090.

Kadu, A., Biswal, B., 2024. A geomorphic perspective on discharge–area relationships. Earth Surf. Process. Landf. 49, 3456–3470. https://doi.org/10.1002/esp.5916.

Kirkby, M.J., 1976. Tests of the random network model, and its application to basin hydrology. Earth Surf. Process. 1, 197–212. https://doi.org/10.1002/esp.3290010302.

Koussis, A.D., Mazi, K., 2016. Reverse flood and pollution routing with the lag-and-route model. Hydrol. Sci. J. 61, 1952–1966. https://doi.org/10.1080/02626667.2015.1061194.

Kouwen, N., Soulis, E.D., Pietroniro, A., Donald, J., Harrington, R.A., 1993. Grouped response units for distributed hydrologic modeling. J. Water Resour. Plann. Manag. 119, 289–305. https://doi.org/10.1061/(ASCE)0733-9496(1993)119:3(289).

Kumar, R., Chatterjee, C., Singh, R.D., Lohani, A.K., Kumar, S., 2007. Runoff estimation for an ungauged catchment using geomorphological instantaneous unit hydrograph (GIUH) models. Hydrol. Process. 21, 1829–1840. https://doi.org/10.1002/hyp.6318.

Leopold, L.B., 1994. A View of the River. Harvard University Press, Cambridge, MA, USA, p. 298.

Leopold, L.B., Maddock Jr., T., 1953. The Hydraulic Geometry of Stream Channels and Some Physiographic Implications. Professional Paper, Washington, D.C. https://doi.org/10.3133/pp252.

Li, J., Zhao, H., Zhang, J., Chen, H., Xu, C.-Y., Li, L., Chen, J., Guo, S., 2020. An improved routing algorithm for a large-scale distributed hydrological model with consideration of underlying surface impact. Hydrol. Res. 51, 834–853. https://doi.org/10.2166/nh.2020.170.

Lohani, A.K., Goel, N.K., Bhatia, K.K.S., 2014. Improving real time flood forecasting using fuzzy inference system. J. Hydrol. 509, 25–41. https://doi.org/10.1016/j.jhydrol.2013.11.021.

López-Vicente, M., Pérez-Bielsa, C., López-Montero, T., Lambán, L.J., Navas, A., 2014. Runoff simulation with eight different flow accumulation algorithms: recommendations using a spatially distributed and open-source model. Environ. Model. Software 62, 11–21. https://doi.org/10.1016/j.envsoft.2014.08.025.

Maidment, D.R., Olivera, F., Calver, A., Eatherall, A., Fraczek, W., 1996. Unit hydrograph derived from a spatially distributed velocity field. Hydrol. Process. 10, 831–844. https://doi.org/10.1002/(SICI)1099-1085(199606.

Manning, R., 1891. On the flow of water in open channels and pipes. Trans. Inst. Civ. Eng. Irel. 20, 161–207.

Markovic, D., Koch, M., 2015. Stream response to precipitation variability: a spectral view based on analysis and modelling of hydrological cycle components. Hydrol. Process. 29, 1806–1816. https://doi.org/10.1002/hyp.10293.

Meyersohn, W.D., 2016. Runoff prediction for dam safety evaluations based on variable time of concentration. J. Hydrol. Eng. 21. https://doi.org/10.1061/(ASCE)HE.1943-5584.0001406.

Mizukami, N., Clark, M.P., Sampson, K., Nijssen, B., Mao, Y., McMillan, H., Viger, R.J., Markstrom, S.L., Hay, L.E., Woods, R., Arnold, J.R., Brekke, L.D., 2016. mizuRoute version 1: a river network routing tool for a continental domain water resources applications. Geosci. Model Dev. (GMD) 9, 2223–2238. https://doi.org/10.5194/gmd-9-2223-2016.

Moody, J.A., Troutman, B.M., 2002. Characterization of the spatial variability of channel morphology. Earth Surf. Process. Landf. 27, 1251–1266. https://doi.org/10.1002/esp.403.

Moore, R.J., 2007. The PDM rainfall-runoff model. Hydrol. Earth Syst. Sci. 11, 483–499. https://doi.org/10.5194/hess-11-483-2007.

Muzik, I., 1996. Flood modelling with Gis-Derived distributed unit hydrographs. Hydrol. Process. 10, 1401–1409. https://doi.org/10.1002/(SICI)1099-1085(199610.

Naden, P.S., 1992. Spatial variability in flood estimation for large catchments: the exploitation of channel network structure. Hydrol. Sci. J. 37, 53–71. https://doi.org/10.1080/02626669209492561.

Nash, J.E., 1957. The form of the instantaneous unit hydrograph. In: International Association of Hydrological Sciences (IAHS) General Assembly, Vol. 3. IAHS Publication, Toronto, Canada, pp. 114–121. No. 45.

Nash, J.E., Sutcliffe, J.V., 1970. River flow forecasting through conceptual models part I — a discussion of principles. J. Hydrol. 10, 282–290. https://doi.org/10.1016/0022-1694(70)90255-6.

Niu, J., Chen, J., Wang, K., Sivakumar, B., 2017. Multi-scale streamflow variability responses to precipitation over the headwater catchments in southern China. J. Hydrol. 551, 14–28. https://doi.org/10.1016/j.jhydrol.2017.05.052.

O'Callaghan, J.F., Mark, D.M., 1984. The extraction of drainage networks from digital elevation data. Comput. Vis. Graph Image Process 28, 323–344. https://doi.org/10.1016/S0734-189X(84)80011-0.

Paul, P.K., Kumari, N., Panigrahi, N., Mishra, A., Singh, R., 2018. Implementation of cell-to-cell routing scheme in a large scale conceptual hydrological model. Environ. Model. Software 101, 23–33. https://doi.org/10.1016/j.envsoft.2017.12.003.

Perrin, C., Michel, C., Andréassian, V., 2003. Improvement of a parsimonious model for streamflow simulation. J. Hydrol. 279, 275–289. https://doi.org/10.1016/S0022-1694(03)00225-7.

Piccolroaz, S., Di Lazzaro, M., Zarlenga, A., Majone, B., Bellin, A., Fiori, A., 2016. HYPERstream: a multi-scale framework for streamflow routing in large-scale hydrological model. Hydrol. Earth Syst. Sci. 20, 2047–2061. https://doi.org/10.5194/hess-20-2047-2016.

Pickup, G., Warner, R.F., 1976. Effects of hydrologic regime on magnitude and frequency of dominant discharge. J. Hydrol. 29. https://doi.org/10.1016/0022-1694(76)90005-6.

Rigon, R., Bancheri, M., Formetta, G., de Lavenne, A., 2016. The geomorphological unit hydrograph from a historical-critical perspective. Earth Surf. Process. Landf. 41, 27–37. https://doi.org/10.1002/esp.3855.

Rinaldo, A., Beven, K.J., Bertuzzo, E., Nicotina, L., Davies, J., Fiori, A., Russo, D., Botter, G., 2011. Catchment travel time distributions and water flow in soils. Water Resour. Res. 47, 1–13. https://doi.org/10.1029/2011WR010478.

Rodríguez-Iturbe, I., González-Sanabria, M., Bras, R.L., 1982. A geomorphoclimatic theory of the instantaneous unit hydrograph. Water Resour. Res. 18, 877–886. https://doi.org/10.1029/WR018i004p00877.

Rodríguez-Iturbe, I., Rinaldo, A., Rigon, R., Bras, R.L., Marani, A., Ijjász-Vásquez, E., 1992. Energy dissipation, runoff production, and the three-dimensional structure of river basins. Water Resour. Res. 28, 1095–1103. https://doi.org/10.1029/91WR03034.

Rodríguez-Iturbe, I., Valdés, J.B., 1979. The geomorphologic structure of hydrologic response. Water Resour. Res. 15, 1409–1420. https://doi.org/10.1029/WR015i006p01409.

Sherman, L.K., 1932. Streamflow from Rainfall by the unit-graph Method.

Solanki, H., Vegad, U., Kushwaha, A., Mishra, V., 2025. Improving streamflow prediction using multiple hydrological models and machine learning methods. Water Resour. Res. 61 (1).

Song, Y., Bindas, T., Shen, C., Ji, H., Knoben, W.J., Lonzarich, L., Clark, M.P., Liu, J., van Werkhoven, K., Lamont, S., Denno, M., 2025. High-resolution national-scale water modeling is enhanced by multiscale differentiable physics-informed machine learning. Water Resour. Res. 61 (4).

Singh, V.P., Yang, C.T., Deng, Z.Q., 2003. Downstream hydraulic geometry relations: 1. Theoretical development. Water Resour. Res. 39. https://doi.org/10.1029/2003WR002484.

Snell, J.D., Sivapalan, M., 1994. On geomorphological dispersion in natural catchments and the geomorphological unit hydrograph. Water Resour. Res. 30, 2311–2323. https://doi.org/10.1029/94WR00537.

Teng, J., Jakeman, A.J., Vaze, J., Croke, B.F.W., Dutta, D., Kim, S., 2017. Flood inundation modelling: a review of methods, recent advances and uncertainty analysis. Environ. Model. Software 90, 201–216. https://doi.org/10.1016/j.envsoft.2017.01.006.

Terink, W., Lutz, A.F., Simons, G.W.H., Immerzeel, W.W., Droogers, P., 2015. SPHY v2.0: spatial processes in HYdrology. Geosci. Model Dev. (GMD) 8, 2009–2034. https://doi.org/10.5194/gmd-8-2009-2015.

Unnithan, S.L.K., Biswal, B., Rüdiger, C., Dubey, A.K., 2024. A novel conceptual flood inundation model for large scale data-scarce regions. Environ. Model. Software 171, 105863. https://doi.org/10.1016/j.envsoft.2023.105863.

Williams, G.P., 1978. Bank-full discharge of rivers. Water Resour. Res. 14. https://doi.org/10.1029/WR014i006p01141.

Zhao, F., Veldkamp, T.I.E., Frieler, K., Schewe, J., Ostberg, S., Willner, S., Schauberger, B., Gosling, S.N., Schmied, H.M., Portmann, F.T., Leng, G., Huang, M., Liu, X., Tang, Q., Hanasaki, N., Biemans, H., Gerten, D., Satoh, Y., Pokhrel, Y., Stacke, T., Ciais, P., Chang, J., Ducharne, A., Guimberteau, M., Wada, Y., Kim, H., Yamazaki, D., 2017. The critical role of the routing scheme in simulating peak river discharge in global hydrological models. Environ. Res. Lett. 12, 075003. https://doi.org/10.1088/1748-9326/aa7250.